

MorningCall: An AI-Powered Adaptive Morning Routine and Productivity System

Authors: Lyton Mhlanga, Lucas Vice, Oseremhen Ewaleifoh, Ann Mathew
Vanderbilt University

Problem & Motivation

Problem

Current productivity tools treat habits, tasks, and routines as separate systems, forcing users to manually coordinate their mornings. These tools are static, fragmented, and increase cognitive load, making it difficult to maintain consistent daily routines. Further, if users desire to keep track of a consistent morning routine, they are forced to manually interact with their phones from the moment they wake up until the very end of their routine.

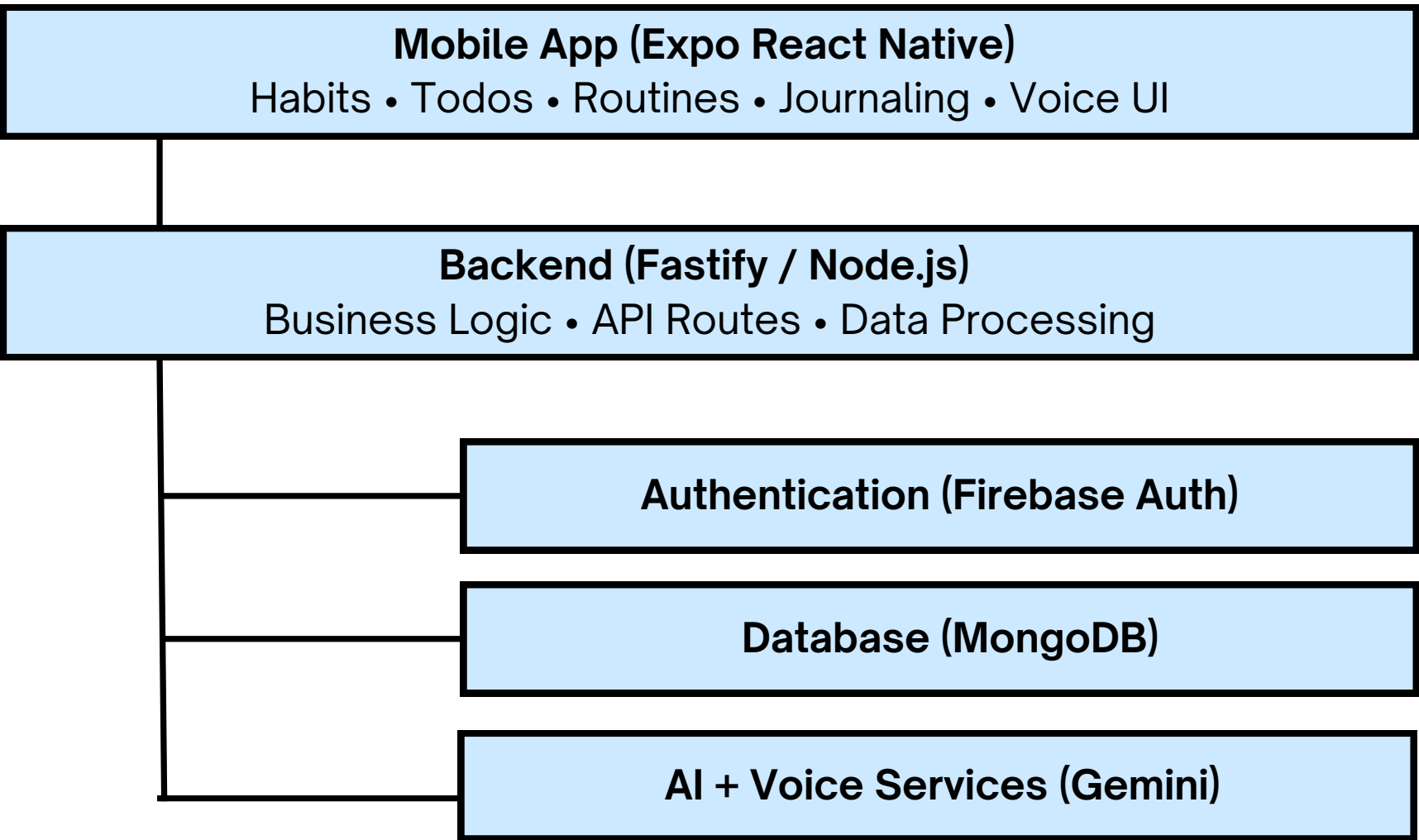
Motivation

Morning routines are critical for long-term productivity and well-being. This project explores how an AI-driven system can unify routine management, reduce friction, and create a more adaptive, hands-free, and context-aware user experience.

This project aims to explore how an intelligent system can:

- Unify habits, tasks, and routines
- Adapt to user needs over time
- Provide an efficient, voice-driven experience

System Architecture

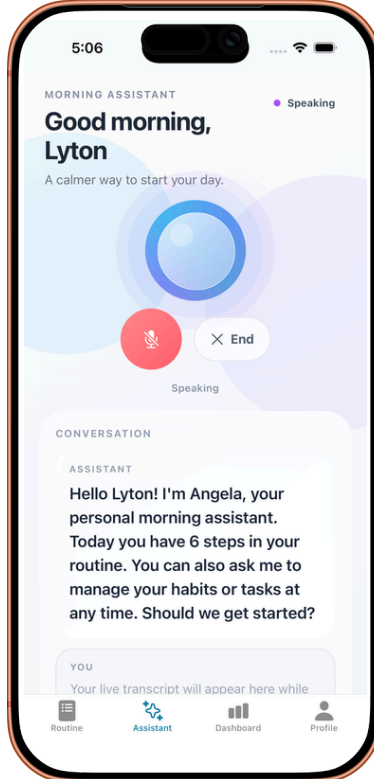


Technical Highlights

MorningCall is designed to move beyond static productivity tools by embedding AI-driven reasoning and interaction directly into the user experience.

What sets our system apart?

- AI Orchestration Layer: Central system manages prompts, context (time, habits, tasks), and conversation flow
- Dynamic Routine Generation: Adapts routines using real-time data instead of fixed workflows
- Voice-First Loop: Seamless input → AI → response → speech interaction
- Stateful Personalization: Learns from user behavior to improve over time



Solution Overview

Solution

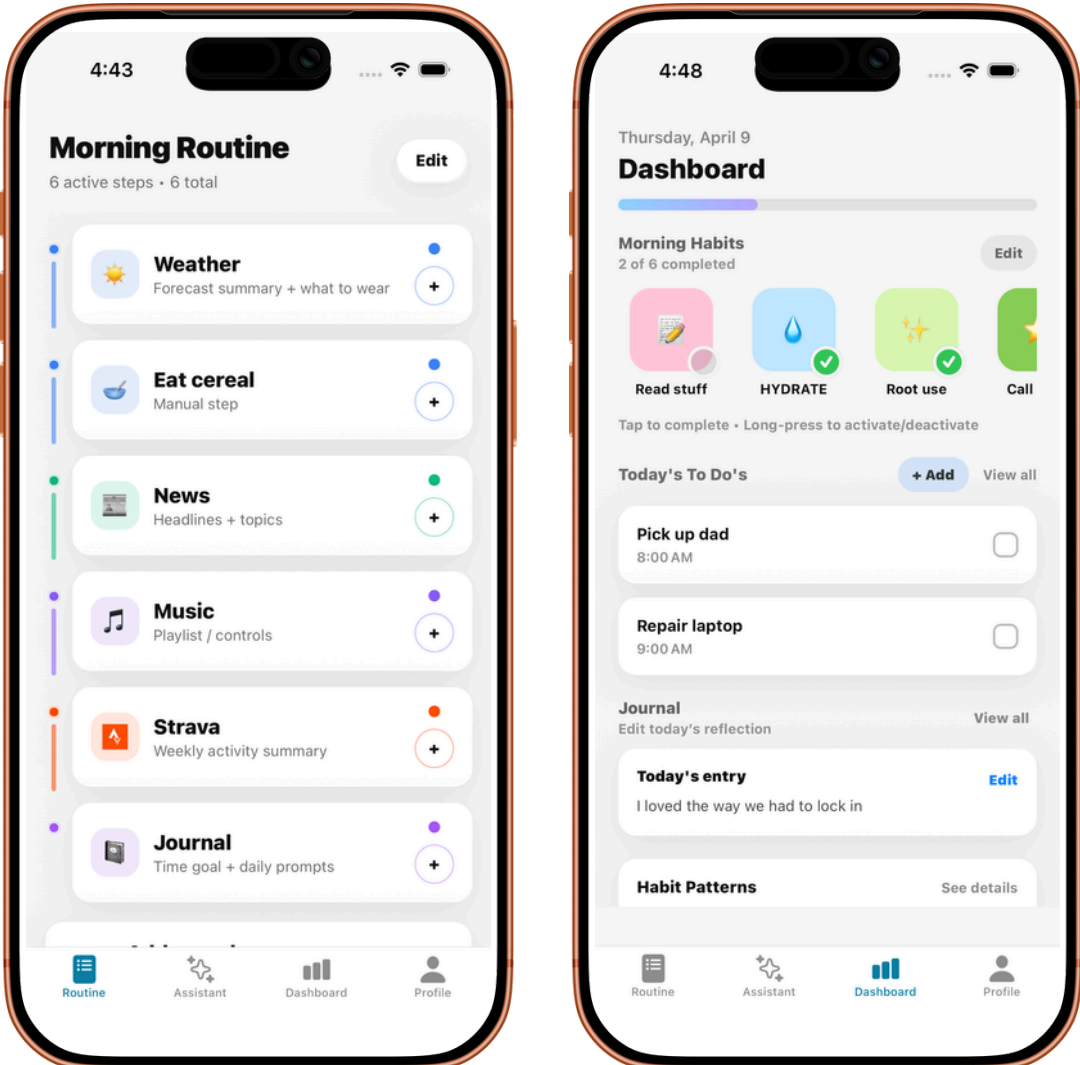
We have designed an AI-powered mobile system to unify habit tracking, task management, and daily routines into a single adaptive and ideally hands-free experience.

The system integrates structured user inputs (habits, todos, routines) with an intelligent backend to support dynamic, context-aware morning planning. By combining multiple productivity workflows into one interface, our app reduces friction and simplifies decision-making during the start of the day.



Key Features

MorningCall provides a unified dashboard that brings together daily habits, prioritized tasks, and routine progress into a single, streamlined interface. Users can track completion, manage tasks, and monitor progress without switching between multiple applications. A customizable routine builder allows flexible creation and reordering of morning steps, enabling users to structure their routines around personal preferences and daily needs.



Challenges & Lessons Learned

Challenges

Integrating AI into a real-time app introduced complexity in managing conversational context and keeping responses relevant. We also faced challenges with state management as habits, tasks, and other user data needed to remain consistent across the system.

Lessons Learned

- Strong, well-defined data models early on prevent major refactoring later when adding adaptive features
- Building AI-first systems requires tight coupling between UX design and system architecture, with continuous iteration
- Regular communication and task tracking (breaking work into small, defined
- Clear Git workflows (branching, PRs) are essential to avoid conflicts and maintain stability pieces) improves cohesion
- Frontend and backend development should happen in parallel